

SPS: Semantically-Guided Prototype Synthesis for Long-Tailed Recognition via Purified Low-Rank Knowledge Transfer

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Abstract—Real-world computer vision recognition tasks often face challenges arising from long-tailed data distributions, where traditional deep learning models tend to be biased toward frequent head classes while underperforming on rare tail classes. This imbalance can be mitigated by transferring knowledge from data-rich head classes to support the representation learning of tail classes; however, such approaches are often susceptible to noise amplification and semantic misalignment. To address these limitations, we introduce SPS, a novel framework that performs semantically-guided knowledge transfer within a purified low-rank subspace. Our framework is built upon two synergistic components: (1) Semantic Low-Rank Fusion (SemLRF) and (2) Prototype Contrastive Loss (ProtoCL). SemLRF constructs a distilled knowledge base by leveraging SVD to project head-class prototypes into a low-rank subspace, effectively capturing shared knowledge while filtering out class-specific noise. Next, guided by the pre-trained CLIP model’s semantic matching, it synthesizes a robust prototype for each tail class by selectively fusing knowledge from the most relevant neighbors within this distilled space. To reinforce feature space discriminability, ProtoCL enforces global separability among all class prototypes. Extensive evaluations on benchmark long-tailed datasets—including CIFAR-10/100-LT, ImageNet-LT, and iNaturalist 2018—demonstrate that our method achieves state-of-the-art performance. Notably, on CIFAR-100-LT (IR=100), SPS boosts a strong baseline to 59.0% overall accuracy, with tail-class accuracy reaching 42.5%, significantly outperforming prior art while maintaining robust head-class performance.

Index Terms—Long-tailed recognition, Knowledge Transfer, Semantic Guidance, Prototype Learning, Low-Rank Subspace

I. INTRODUCTION

While deep learning has achieved remarkable progress in various vision tasks, its performance heavily relies on balanced data distributions, a condition rarely met in the real world. Real-world data inherently exhibits a long-tailed nature: a few head classes contain an abundance of samples, while a vast number of tail classes are severely underrepresented. For example, in autonomous driving scenarios, common objects like vehicles and pedestrians constitute the head classes, whereas rare traffic signs or specific obstacle types form the tail classes. This inherent class imbalance induces a strong learning bias in conventional deep learning models. When

trained on imbalanced data, these models suffer a dramatic performance degradation on tail classes, with accuracy drops often exceeding 30% [1]. This degradation significantly undermines the reliability and generalization of models in applications such as medical diagnosis, industrial defect detection, and autonomous driving perception. Consequently, effectively mitigating the long-tailed effect has become a critical bottleneck for deploying deep learning systems beyond controlled laboratory settings.

To address long-tailed distribution challenges, researchers have proposed various strategies, broadly categorized into three groups [2]: rebalancing [3], [4], information enhancement [5], [6], and module improvement [7]–[9]. Among them, the category rebalancing method directly addresses the issue of data imbalance by making adjustments at the data [4] and loss function levels [3]. In contrast, the module improvement methods enhance the model’s performance in handling long-tailed distributions by optimizing the network structure [9] and improving the training strategy [7]. Within the information enhancement category, the paradigm of knowledge transfer has emerged as particularly promising, as it seeks to leverage rich knowledge from head classes to bolster tail class representations [6], [10], [11]. Although these methods aim to reduce the performance disparity through feature transfer or knowledge distillation, they still face fundamental challenges in effectively transferring knowledge from head to tail classes.

Existing knowledge transfer methods face two key challenges. The first challenge is related to *knowledge purity*. Directly transferring features from the head classes introduces significant instance-level variance and noise irrelevant to the class’s semantic essence. As illustrated in Figure 1(a), this unrefined knowledge transfer can contaminate the sensitive feature learning process of the data-scarce tail classes (e.g., “fox”). Consequently, the learned feature space, indicated by the shaded gray region, becomes structurally biased and misaligned, causing poor generalization and failure to encompass the majority of validation samples (represented by green triangles). The second challenge is related to *knowledge source breadth*. Current strategies often oscillate between two suboptimal extremes. On one hand, indiscriminately aggregating all head classes can induce negative transfer across wide semantic gaps. On the other hand, relying solely on the most similar head class often leads to overfitting and diminished generalization. As Figure 1(a) illustrates, this latter approach results in an excessively narrow feature distribution for “fox,” which is indicative of poor generalization.

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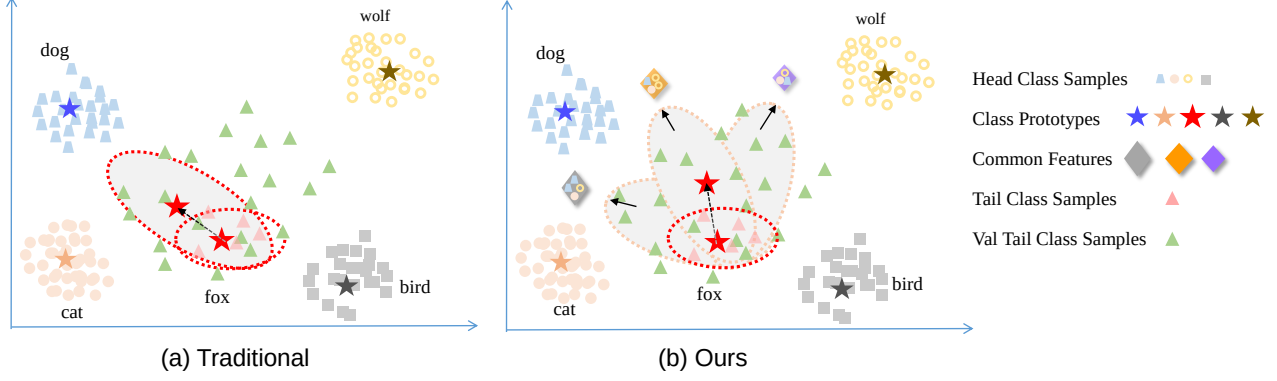


Fig. 1: Motivation of our proposed method. (a) Traditional methods directly fuse head-class features, creating a biased feature space. (b) Our method, in contrast, first distills head-class knowledge into a shared, low-rank subspace (represented by the diamonds) and leverages it to synthesize a robust and purified guidance prototype for the tail class, ultimately constructing a better-structured and less-biased feature space.

This dilemma is particularly critical in real-world long-tailed recognition, where tail classes contain very few samples and their prototypes are highly sensitive to sampling noise. Although transferring knowledge from data-rich head classes can enrich tail-class representations, overly broad transfer may introduce irrelevant head-class variations and cause negative transfer, whereas overly narrow transfer may lead to overfitting to a limited semantic neighborhood. Standard head-to-tail transfer methods mainly increase the amount of transferred information, but they do not explicitly distinguish transferable common semantics from class-specific or instance-specific noise. In contrast, prototype or contrastive learning improves inter-class separability, but it does not determine which head classes should provide guidance or what knowledge should be transferred. Therefore, an effective solution should jointly purify transferable head-class knowledge, select semantically appropriate source classes, and maintain global prototype discriminability.

Motivated by the above analysis, we propose to synthesize tail-class prototypes from purified and semantically relevant head-class knowledge. The key idea is to first distill generalizable common semantics from head-class prototypes and then use them as reliable guidance for tail-class prototype learning. As illustrated in Figure 1(b), our approach first distills pure, generalizable *common features* (represented by diamonds) from a diverse set of semantically relevant head classes (e.g., “dog,” “cat,” and “wolf”). Next, the tail class prototype is enhanced by incorporating the distilled general features, improving the overall category representation. As can be seen in Figure 1(b), the feature space of “fox” is expanded by learning from the general features. As a result, the prototype of the tail class (the red star) moves to a more reasonable position in the feature space, the feature space of “fox” is significantly expanded to cover more validation samples, thereby resolving the purity-breadth problems and achieving superior generalization performance.

Specifically, we introduce a novel training framework Semantically-Guided Prototype Synthesis (SPS) to address the

“purity–breadth dilemma”. First, to address the issue of knowledge purity, we construct a shared low-rank Purified Visual Subspace (PVS) by applying Singular Value Decomposition (SVD) to head-class prototypes. Second, to tackle the problem of knowledge source breadth, we identify semantically relevant head classes for each tail class through label semantics captured by a pre-trained CLIP model [12]. Third, within the PVS, a guidance prototype is synthesized via weighted fusion of these relevant head classes. The SemLRF loss then encourages the tail-class prototype to align with this semantically-guided target. Finally, to ensure global discriminability and prevent prototype collapse, we introduce a Prototype Contrastive Loss (ProtoCL) that maximizes the separability among all class prototypes, reinforcing the structure of the learned feature space.

Comprehensive experiments demonstrate that the proposed SPS framework outperforms recent state-of-the-art methods in long-tailed learning tasks. Specifically, on CIFAR-10-LT (IR=100), CIFAR-100-LT (IR=100), ImageNet-LT, and iNaturalist 2018, we achieved accuracies of 87.6%, 59.0%, 60.7%, and 75.6%, respectively.

Our main contributions are summarized as follows:

- We address the purity-breadth dilemma in long-tailed transfer learning via Semantic Low-Rank Fusion—a novel method that improves both knowledge purity and breadth through semantic-guided transfer.
- We propose a Semantically-Guided Prototype Synthesis framework for long-tailed recognition, including Semantic Low-Rank Fusion to obtain the synthetic prototypes and Prototype Contrastive Loss to achieve global discriminability.
- We conduct extensive experiments to validate that our proposed method effectively improves tail-class accuracy without compromising head-class accuracy.

II. RELATED WORKS

A. Long tail learning

Long-tailed learning aims to address the ubiquitous class imbalance problem found in real-world datasets, where a few head classes are data-rich while most tail classes are data-scarce. This distribution is widespread in tasks such as face recognition [8], medical image diagnosis [13], and remote sensing classification [14]. Such an imbalance causes conventional deep learning models to become heavily biased towards the well-represented head classes, leading to a significant decline in recognition accuracy for tail classes [8], [13], [14].

Existing strategies to mitigate this issue can be broadly categorized. At the data level, a common approach involves rebalancing the data distribution. This includes methods like oversampling rare class instances [15] and sophisticated data augmentation techniques [4], [9], both aimed at alleviating the model bias caused by data scarcity. At the model and loss level, another line of work focuses on adapting the learning process. These methods include reweighting strategies that adjust the influence of samples by modifying class weights in the loss function [16] or rescaling logits [17]. Further innovations in this area involve designing specialized modules [10] or novel loss functions [3] to enhance the representations of tail classes specifically. However, a fundamental limitation shared by these approaches is that they primarily address the problem from a statistical standpoint, focusing on class frequencies while largely treating each class as an isolated entity. Thus, they cannot exploit the fact that a data-poor tail class like “fox” could borrow knowledge from a data-rich and semantically close head class like “dog,” a limitation that motivates the shift to explicit knowledge transfer pursued in this work.

Recent studies have further extended long-tailed learning from more diverse perspectives. Pipeline-oriented analyses summarize long-tailed learning as a systematic process involving data, representation, classifier, and training-strategy design [18]. Boundary- and geometry-oriented methods improve long-tailed recognition by refining decision regions or reshaping feature distributions. For example, category-information-based methods adjust decision boundaries according to class learning difficulty [19], curvature-balanced feature manifold learning calibrates the geometry of feature manifolds [20], and Gaussian clouded logit adjustment enlarges the effective decision space of tail classes through class-dependent logit perturbation [21]. Another line of work exploits majority-class context or pretrained representations, including context-rich minority oversampling [22], lightweight fine-tuning with foundation models [23], and Gaussian neighborhood minimization for visual prompt tuning [24]. Although these methods improve long-tailed recognition from data augmentation, decision-boundary refinement, feature-geometry calibration, or foundation-model adaptation perspectives, they do not explicitly address how to transfer head-class knowledge to tail classes while jointly controlling knowledge purity and source breadth.

B. Knowledge Transfer

Knowledge transfer plays a crucial role in long-tailed learning [6], [10], [25]–[28]. The core idea is to leverage rich head-class knowledge to enhance tail-class representations, which is implemented in existing methods through various strategies. One common approach is parameter sharing, where a backbone network is jointly trained to learn generalizable features; for instance, the two-stage BBN [29] method utilizes a shared feature extractor to benefit from data-rich head classes. Another popular strategy is knowledge distillation, exemplified by GML [10], which uses knowledge from a teacher model to provide high-quality contrastive samples for tail classes. Recent methods further explore knowledge excavation and calibration under long-tailed distributions. For example, SHIKE [27] integrates heterogeneous experts with knowledge excavation to improve representation learning, while uncertainty calibration transfers distributional statistics from head classes to tail classes for better calibrated predictions [28]. LCReg [11] learns a set of class-agnostic latent categories/features and uses their similarity maps with image features to re-represent object features. A more direct method is feature transfer, such as the H2T [6] approach, which fuses head-class features directly into tail-class representations.

However, despite these varied strategies, existing methods often face the inherent “purity–breadth dilemma.” Direct instance-level transfer can introduce noise, compromising representation purity, while selecting optimal knowledge sources remains challenging. For example, H2T [6] directly transfers head-class features to tail classes, but such feature-level fusion may inevitably introduce class-specific noise from head classes. LCReg [11] learns shared latent categories to re-represent object features, but it focuses more on latent feature reconstruction than on explicitly identifying which head-class knowledge is clean and semantically suitable for tail classes. In contrast, SPS performs prototype-level transfer after low-rank purification and semantic neighbor selection. Therefore, our method does not simply increase the amount of transferred head-class information, but explicitly controls both the purity of transferred knowledge and the semantic breadth of source classes.

C. Prototype Learning

Prototype learning, which represents each class with a central vector or “prototype,” is a foundational concept in various domains like few-shot and long-tailed learning [30]–[32]. In long-tailed scenarios, prototypes are frequently employed to enhance the feature consistency and noise robustness of tail classes. For instance, PaCo [33] introduces class centers as contrastive learning targets; SBCL [34] leverages subclass prototypes to improve discriminability; NLT [32] integrates optimal transport to construct cleaner representations by refining class prototypes; and BCL [35] extracts classifier weights as prototypes to mitigate the bias of contrastive learning in long-tailed settings. Existing methods treat prototypes passively, often failing to optimize tail-class prototypes effectively. In contrast, our approach actively refines these prototypes by

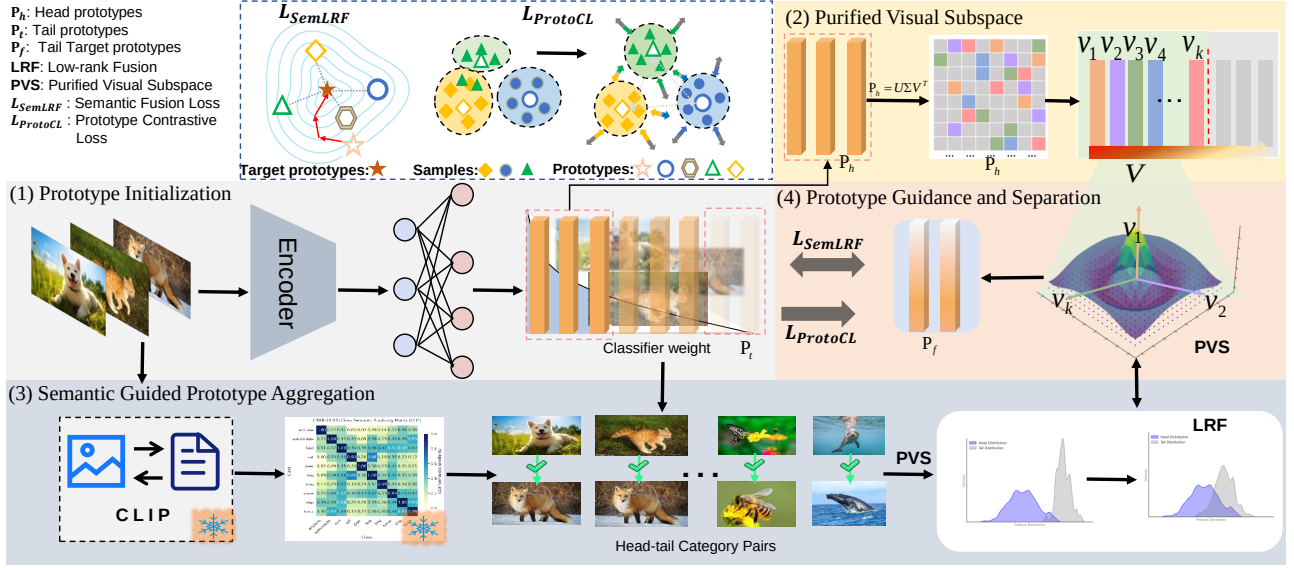


Fig. 2: An overview of our Semantically Guided Prototype Synthesis (SPS) framework. The method consists of four stages: (1) Initializing category prototypes from classifier weights; (2) Constructing a purified visual subspace based on rich head-class knowledge; (3) Within this subspace, CLIP-guided head-tail pair semantic matching is employed to facilitate the synthesis of enhanced tail-class prototypes; and (4) Finally, Semantic Low-Rank Fusion Loss (SemLRF) and Prototype Contrastive Loss (ProtoCL) are introduced to enhance the representation of tail-class prototypes and promote clear separation among all class prototypes.

incorporating distilled head-class knowledge, significantly improving their quality and performance.

III. METHODS

A. Overall Framework

SPS framework addresses the “purity-breadth” dilemma in long-tailed recognition by reshaping the feature space, enhancing robustness and discriminability, especially for tail classes. As shown in Figure 2, the framework first extracts features from input images and initializes class prototypes using the classifier’s weight matrix. Next, the model performs low-rank decomposition to construct a purified visual subspace using head class prototypes. This subspace is then employed to generate refined guidance prototypes for the tail classes. Moreover, the framework synthesizes semantic guidance prototypes using a pre-trained CLIP model, which matches head and tail class pairs based on semantic similarity. These matched prototypes are then projected into the purified visual subspace (PVS) for weighted fusion, after which they are mapped back to the original feature space to align with the global structure. Finally, Semantic Low-Rank Fusion Loss and Prototype Contrastive Loss are applied to optimize the tail class prototypes, enhancing overall classification performance.

These components are designed to work in a coordinated manner. SemLRF purifies head-class knowledge by projecting prototypes into a low-rank visual subspace, while CLIP-guided semantic synthesis selects relevant head classes and fuses their purified prototypes to provide stable guidance for tail classes. ProtoCL further enforces global prototype separability, preventing the synthesized tail prototypes from collapsing toward head classes and improving overall discriminability.

B. Prototype Initialization

To capture rich features and establish a direct link to class representations, our model employs a feature extractor, $f_\theta(\cdot)$, followed by a linear classifier. For a given input image $\mathbf{x}$, the feature extractor (e.g., a ResNet backbone) maps it to a d -dimensional feature vector, $\mathbf{z} = f_\theta(\mathbf{x})$, where $\mathbf{z} \in \mathbb{R}^{1 \times d}$. We leverage the classifier-as-prototype paradigm [35], where the weight vectors of the classifier are directly treated as the class prototypes. The weight matrix of this linear classifier is denoted as $\mathbf{W} \in \mathbb{R}^{C \times d}$, where C is the total number of classes and d is the feature dimension.

Crucially, we define each row vector $\mathbf{w}_c \in \mathbb{R}^{1 \times d}$ of the weight matrix $\mathbf{W}$ as the prototype $\mathbf{p}_c$ for class c . We therefore have a set of prototypes $\{\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_C\}$, where $\mathbf{p}_c = \mathbf{w}_c$. Under this definition, the classification process can be viewed as computing the dot-product similarity between the input feature $\mathbf{z}$ and every class prototype. The final classification logits are obtained by the matrix multiplication:

$$\text{logits} = \mathbf{z}\mathbf{W}^\top. \quad (1)$$

This operation is equivalent to taking the dot product of the feature vector $\mathbf{z}$ with each prototype (row of $\mathbf{W}$). This “weight-as-prototype” paradigm allows us to directly analyze and manipulate the classifier weights, providing a solid foundation for the subsequent knowledge transfer and fusion mechanisms.

C. Distilling a Purified Visual Subspace

A core challenge in long-tailed learning is that directly transferring knowledge from data-rich head classes can introduce instance-specific noise and bias. This is because

head class prototypes, while robust, are inevitably shaped by the idiosyncrasies of their many training samples, rather than representing pure, abstract semantics. To overcome this, we hypothesize that the pure, transferable visual semantics shared among head classes do not occupy the entire feature space, but are instead encapsulated within a compact, low-rank subspace. Our approach is therefore to identify and isolate this core subspace by analyzing the principal components of the head-class prototypes. By projecting onto this subspace, we effectively filter out instance-specific noise and idiosyncrasies from the head-class samples, thus forming a purified space that represents core, transferable knowledge.

In detail, following the above thoughts, we begin by assembling the prototypes of the H head classes into a matrix $\mathbf{P}_h \in \mathbb{R}^{H \times d}$. To extract the underlying shared visual structure from the head class prototypes, we apply Singular Value Decomposition (SVD) to this matrix:

$$\mathbf{P}_h = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top, \quad (2)$$

where $\mathbf{U} \in \mathbb{R}^{H \times H}$ contains the left singular vectors, $\mathbf{\Sigma} \in \mathbb{R}^{H \times d}$ is a diagonal matrix of singular values, and $\mathbf{V} \in \mathbb{R}^{d \times d}$ contains the right singular vectors.

The columns of $\mathbf{V}$ form an orthonormal basis for the d -dimensional feature space, ordered by the amount of variance they capture within the head-class prototypes. These axes may correspond to high-level concepts (e.g., an axis for “furry textures” or “metallic surfaces”) that are common to multiple object categories. By projecting features onto the subspace defined by these top vectors, we can filter out instance-specific noise and retain the core, transferable knowledge.

Therefore, we construct our low-rank **Purified Visual Subspace** (PVS) by forming a projection matrix $\mathbf{V}_k \in \mathbb{R}^{d \times k}$ from the **top- k right singular vectors** of $\mathbf{V}$ corresponding to the k largest singular values:

$$\mathbf{V}_k = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k] \in \mathbb{R}^{d \times k}. \quad (3)$$

This matrix $\mathbf{V}_k$ serves as our projection basis. Any prototype projected into this subspace is effectively distilled, retaining its shared semantic essence while discarding noise.

D. External Semantic Guided Prototype Aggregation for Tail Classes

In long-tailed classification, the representations of tail classes are scarce, which leads to the imbalanced learning of the model. To enhance the representations of tail classes, we propose a mechanism to synthesize a new, more robust prototype for each tail class. This synthesis process, illustrated in Figure 3, fuses the tail class’s own information with knowledge transferred from a select group of semantically similar head classes. The entire procedure is guided by two key components: a fixed semantic prior from CLIP and a data-driven Purified Visual Subspace (PVS).

The synthesis of the fused prototype $\mathbf{p}'_f$ for a given tail class prototype $\mathbf{p}_t$ is performed as follows:

a) *Identification of Semantic Neighbors*: To effectively guide tail class learning, we first identify head classes that are semantically similar to the target tail class t . Using our pre-computed CLIP semantic similarity matrix, we identify the N head classes with the highest similarity scores to class t . This yields a small set of relevant head prototypes, $\mathbf{P}_h^{\text{topN}} = \{\mathbf{p}_{h_1}, \dots, \mathbf{p}_{h_N}\}$, and their corresponding similarity scores. These scores are then normalized via a Softmax function to produce a set of fusion weights, $\mathbf{w} \in \mathbb{R}^N$.

b) *Projection into the Purified Visual Subspace*: To eliminate instance-specific noise and retain only the shared, essential visual semantics, we project the relevant prototypes into the more compact and stable k -dimensional Purified Visual Subspace (PVS), which is defined by the basis $\mathbf{V}_k$. Both the tail prototype and the selected head prototypes are transformed into their low-rank representations:

$$\mathbf{p}'_t = \mathbf{p}_t \mathbf{V}_k, \quad (4)$$

$$\mathbf{p}'_{h_i} = \mathbf{p}_{h_i} \mathbf{V}_k, \quad \forall \mathbf{p}_{h_i} \in \mathbf{P}_h^{\text{topN}}. \quad (5)$$

where $\mathbf{p}' \in \mathbb{R}^{1 \times k}$ denotes a low-rank class prototype in the PVS.

c) *Guided Fusion in the Subspace*: Within the Purified Visual Subspace (PVS), we synthesize a guidance prototype, $\mathbf{p}'_{\text{guide}}$, to serve as a purified semantic reference for the tail class. This is achieved by computing the weighted average of the low-rank head-class prototypes, based on the similarity weights $\mathbf{w}$ obtained in Step 1.

$$\mathbf{p}'_{\text{guide}} = \sum_{i=1}^N w_i \cdot \mathbf{p}'_{h_i}, \quad (6)$$

This guidance prototype should be a collection of shared visual semantics of the relevant head classes. Its purpose is to guide the learning of the tail class prototype towards a more generalizable representation, inferred from its data-rich neighbors. The final fused low-rank representation, $\mathbf{p}'_f$, is then formed by a convex combination of the tail’s low-rank representation and this guidance vector, controlled by a balancing hyperparameter w_{tail} :

$$\mathbf{p}'_f = (1 - w_{\text{tail}})\mathbf{p}'_t + w_{\text{tail}}\mathbf{p}'_{\text{guide}}. \quad (7)$$

The resulting Low-rank prototype, $\mathbf{p}'_f$, serves as a semantically aligned and structurally stable target for guiding the tail class prototype. It effectively integrates original knowledge from the tail class and shared structural knowledge distilled from the head classes, enabling more robust prototype learning in the subsequent optimization.

E. Prototype Guidance and Separation

While the standard classification loss ($\mathcal{L}_{\text{cls}}$) provides a basic foundation, it struggles to learn a robust feature space under severe class imbalance. Its sample-driven nature causes two major issues: unreliable tail-class prototypes due to data scarcity, and a biased feature space dominated by head-class variance, which compresses tail classes and blurs their decision boundaries. To address these challenges, we propose two complementary strategies. **Semantic Low-Rank Fusion Loss**

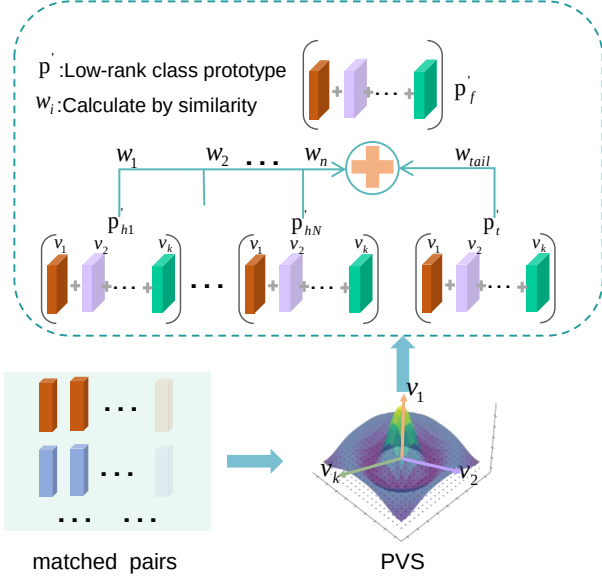


Fig. 3: Illustration of our Semantic-Guided Fusion process.

enhances tail-class prototypes by integrating knowledge from head classes, mitigating weak tail representations. **Prototype Contrastive Loss** promotes global discriminability by enforcing clearer separation among all class prototypes. Together, these methods guide the model to learn a more balanced and robust feature space.

a) Semantic Low-Rank Fusion Loss ($\mathcal{L}_{\text{SemLRF}}$): To address the noisy, data-scarce prototypes of tail classes, we introduce the Semantic Low-Rank Fusion Loss. This regularization guides the tail prototypes toward a more stable and rich representation by leveraging knowledge from semantically similar head classes. As outlined in the previous section, we first synthesize a fused target prototype, $\mathbf{p}'_f$, for each tail class $t \in \mathcal{C}_{\text{tail}}$ by blending the tail class's information with knowledge from its closest head-class neighbors within a dynamic Purified Visual Subspace (PVS).

Once this fused prototype is obtained, it is projected back to the original d -dimensional feature space, resulting in the final synthesized prototype $\mathbf{p}_f$:

$$\mathbf{p}_f = \mathbf{p}'_f \mathbf{V}_k^\top, \quad (8)$$

To enforce this fusion, we define the regularization term as the Mean Squared Error (MSE) between the current tail prototype $\mathbf{p}_t$ and the synthesized target $\mathbf{p}_f$:

$$\mathcal{L}_{\text{SemLRF}} = \|\mathbf{p}_t - \mathbf{p}_f\|_2^2 \quad (9)$$

The target prototype $\mathbf{p}_f$ is detached from the computation graph to prevent gradients from flowing back through the synthesis process. The fusion loss is then computed as the average of these individual losses over all tail classes. Finally, this regularization is activated after an initial warmup period to allow the PVS to stabilize before guiding the tail prototypes. The complete implementation process can be found in supplementary materials.

b) Prototype Contrastive Loss ($\mathcal{L}_{\text{ProtoCL}}$): To enhance the overall discriminability of the feature space, we employ a contrastive loss that explicitly pushes all class prototypes apart from each other. We formulate this using the InfoNCE objective, where for each class prototype $\mathbf{p}_i$, it is treated as an anchor, and all other prototypes $\{\mathbf{p}_j\}_{j \neq i}$ are treated as negatives.

After L2-normalizing all prototypes, we compute a pairwise cosine similarity matrix $\mathbf{S}$. The separation loss for a single prototype $\mathbf{p}_i$ is then:

$$\mathcal{L}_{\text{ProtoCL}} = -\log \frac{\exp(S_{ii}/\tau)}{\sum_{j=1}^C \exp(S_{ij}/\tau)}, \quad (10)$$

where τ is the temperature parameter (set to 1 in our implementation). The total separation loss, $\mathcal{L}_{\text{ProtoCL}}$, is the average of these individual losses over all C classes. Crucially, this loss is applied with a delayed activation strategy; it is only enabled after the model has trained for a substantial number of epochs (e.g., half of the total training epochs), once the prototypes have achieved initial stability.

c) Overall Training Objective.: The final loss function for our model is a weighted combination of the main classification loss ($\mathcal{L}_{\text{cls}}$) and our two regularization terms. The complete training objective is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \lambda_s \cdot \mathcal{L}_{\text{SemLRF}} + \lambda_p \cdot \mathcal{L}_{\text{ProtoCL}}. \quad (11)$$

where λ_s and λ_p are hyperparameters that control the strength of each regularization. Note that λ_p is effectively zero during the initial training phase, consistent with its delayed activation strategy.

IV. EXPERIMENTS

A. Experiments Setting

Datasets We used CIFAR-10/100-LT, ImageNet-LT, and iNaturalist 2018 as the benchmark datasets. The specific datasets can be obtained in the supplementary materials.

Hyperparameters for Our Method (SPS). We set the loss weights to $\lambda_s = 10$ and $\lambda_p = 2$, balancing the contributions of the proposed regularization terms. For the SemLRF module, we configure the rank as $k = 16$, select the top $N = 5$ most similar head classes for semantic transfer, and apply a fusion weight of $w_{\text{tail}} = 0.5$ to control the degree of tail-class enrichment.

B. Results on CIFAR-LT

1) Overall Performance on CIFAR-LT: To comprehensively evaluate the effectiveness of our proposed SPS framework, we conduct extensive comparisons against a range of state-of-the-art methods [1], [5], [6], [9], [17], [35], [36], [38]–[43] on the CIFAR-10-LT and CIFAR-100-LT benchmarks.

First, our method demonstrates superior performance, especially in challenging scenarios. On CIFAR-100-LT with an imbalance ratio of 100, our GLMC+SPS outperforms both GLMC and the recent DBM-NCL, achieving an accuracy improvement of 0.8% and 1.5%, respectively. This validates our core approach of synthesizing guidance for tail classes

TABLE I: Top-1 Accuracy (%) on CIFAR-10-LT and CIFAR-100-LT under imbalance ratios (IR) of 10, 50, and 100. We integrate our proposed SPS method with multiple baseline approaches.

Method	Venue	CIFAR-10-LT			CIFAR-100-LT		
		IR=10	IR=50	IR=100	IR=10	IR=50	IR=100
RIDE [36]	ICLR'21	—	—	—	—	—	48.0
BCL [35]	CVPR'22	91.1	87.2	84.3	64.9	56.6	51.9
WD [17]	CVPR'22	91.9	85.7	82.4	68.3	57.1	52.2
GLMC [9]	CVPR'23	<u>94.5</u>	<u>90.5</u>	86.3	<u>72.7</u>	<u>62.0</u>	<u>58.2</u>
ECL [37]	TMM'23	91.8	88.9	86.5	67.3	59.9	56.3
DLLA [38]	TCSVT'24	91.3	87.3	85.3	63.8	54.8	50.1
RIDE+H2T [6]	AAAI'24	—	—	—	—	55.5	51.4
ConCutMix [5]	TIP'24	91.4	88.0	86.1	64.5	57.4	53.2
Proco [1]	TPAMI'24	91.9	88.2	85.9	65.5	57.1	52.8
PADE [30]	TMM'24	—	—	—	—	54.3	50.8
FUR [39]	IJCV'24	90.9	86.2	83.7	61.8	54.1	50.9
DBM-BCL [40]	AAAI'25	91.7	87.2	84.6	65.3	56.0	51.7
DBM-NCL [40]	AAAI'25	93.2	89.9	<u>87.5</u>	69.8	<u>62.0</u>	57.5
LOS [41]	ICLR'25	92.4	86.1	83.5	68.5	57.8	53.4
FeatRecon [42]	ICLR'25	91.6	87.8	85.2	65.3	57.0	52.5
WD+SPS	Ours	92.0	86.1	82.7	68.7	57.3	52.7
LOS+SPS	Ours	92.3	86.7	83.8	68.9	58.0	54.0
GLMC+SPS	Ours	94.6	90.7	87.6	73.5	62.9	59.0

TABLE II: Top-1 Accuracy (%) on CIFAR-100-LT (IR=100), evaluated on head, medium, and tail class subsets.

Method	Venue	Head	Mid	Tail	Acc
RIDE [36]	ICLR'21	68.1	49.2	23.9	48.0
BCL [35]	CVPR'22	67.2	53.1	32.9	51.9
WD [17]	CVPR'22	70.5	48.9	34.7	52.2
GLMC [9]	CVPR'23	73.9	60.4	37.9	<u>58.2</u>
RIDE+H2T [6]	AAAI'24	—	—	—	51.4
ConCutMix [5]	TIP'24	—	—	—	53.2
Proco [1]	TPAMI'24	69.0	52.7	34.1	52.8
FUR [39]	IJCV'24	—	—	—	50.9
LCSL [43]	TCSVT'24	78.6	53.0	13.4	51.0
DBM-BCL [40]	AAAI'25	67.9	51.9	32.4	51.7
DBM-NCL [40]	AAAI'25	71.5	<u>59.1</u>	<u>39.3</u>	57.5
LOS [41]	ICLR'25	71.3	52.6	33.4	53.4
FeatRecon [42]	ICLR'25	68.2	53.3	34.0	52.5
WD+SPS	Ours	69.9	49.7	36.1	52.7
LOS+SPS	Ours	71.8	51.9	35.5	54.0
GLMC+SPS	Ours	<u>74.1</u>	58.7	42.5	59.0

TABLE III: Top-1 Accuracy (%) on ImageNet-LT and iNaturalist 2018 under long-tailed settings.

Method	Venue	ImageNet-LT	iNaturalist
RIDE [36]	ICLR'21	54.9	71.4
ResLT [44]	TPAMI'22	52.7	70.7
WD [17]	CVPR'22	53.9	70.2
BCL [35]	CVPR'22	56.7	71.3
DRW+OTmix [45]	NeurIPS'23	53.4	71.1
GLMC [9]	CVPR'23	54.9	70.2
DLLA [38]	TCSVT'24	54.5	71.3
DSCL [7]	AAAI'24	57.7	72.0
ResLT+H2T [6]	AAAI'24	53.4	71.9
IWB [16]	AAAI'24	55.2	71.5
LCSL [43]	TCSVT'24	55.4	—
DBM-BCL [40]	AAAI'25	56.3	71.9
DBM-GML [40]	AAAI'25	57.4	72.0
CBTS [46]	TMM'25	<u>60.6</u>	<u>75.4</u>
LOS [41]	ICLR'25	54.4	70.8
GLMC+SPS	Ours	55.4	72.1
BCL+SPS	Ours	58.2	71.9
CBTS+SPS	Ours	60.7	75.6

within a purified, low-rank semantic subspace. Additionally, our framework shows strong robustness across varying imbalance ratios, consistently maintaining a competitive edge even as data scarcity increases from IR 10 to 100. This indicates that our semantic-guided knowledge fusion is a stable and effective strategy for long-tailed recognition. Finally, SPS proves to be a complementary module, enhancing its performance when integrated with approaches like WD and LOS by focusing on semantic prototypes.

2) *Detailed Analysis on CIFAR-100-LT*: To gain deeper insight into how our method improves performance, we present a fine-grained analysis on the Head, Medium, and Tail class subsets for CIFAR-100-LT (IR=100) in Table II. The results reveal two crucial advantages of our approach:

First, our SPS framework significantly boosts tail-class accuracy, achieving a 4.6% absolute gain when applied to GLMC, validating the effectiveness of SemLRF and ProtoCL in transferring purified knowledge and enhancing discrimina-

tive boundaries. Notably, this improvement in tail-class accuracy is achieved without sacrificing head-class performance. In fact, head accuracy slightly increases for strong baselines like GLMC and LOS, suggesting that our approach enriches tail-class representations while maintaining the structure of head-class features. Finally, a slight performance drop is observed for medium-frequency classes, likely due to the feature space reorganization caused by the combination of SemLRF and ProtoCL. The trade-off between local geometry (tail class enrichment) and global alignment (class margin stabilization) might lead to this slight performance drop for medium-frequency classes.

C. Results on Large-Scale Datasets

1) *Overall Performance*: To verify the scalability and generalization capability of our framework beyond smaller-scale

benchmarks [6], [7], [9], [16], [17], [35], [36], [38], [40], [41], [43]–[46], we conduct extensive experiments on the large-scale ImageNet-LT and iNaturalist 2018 datasets.

As shown in Table III, our method demonstrates strong and scalable performance on challenging benchmarks. Most notably, SPS serves as a powerful plug-and-play module that significantly boosts existing baselines. For instance, integrating SPS with BCL on ImageNet-LT yields 58.2% accuracy—an improvement of +1.5% over BCL and surpassing recent methods like DSCL and DBM-GML, highlighting its complementarity to contrastive learning approaches. On iNaturalist 2018, SPS also remains compatible with strong recent baselines. When integrated with CBTS, SPS improves the overall accuracy from 75.4% to 75.6%. Although the gain is modest, this result is obtained on a highly competitive baseline. This suggests that SPS can still provide additional benefits to underrepresented classes even when the baseline performance is already strong.

TABLE IV: Top-1 Accuracy (%) on ImageNet-LT, reported for head, medium, and tail class groups as well as the overall average (Acc).

Method	Venue	Head	Mid	Tail	Acc
RIDE [36]	ICLR’21	66.2	51.7	34.9	54.9
BCL [35]	CVPR’22	67.2	53.9	36.5	56.7
GLMC [9]	CVPR’23	66.8	52.1	29.8	54.9
DRW+OTmix [45]	NeurIPS’23	67.0	49.0	30.4	53.4
DLLA [38]	TCSVT’24	–	–	–	54.5
DSCL [7]	AAAI’24	68.5	55.2	35.4	57.7
DBM-BCL [40]	AAAI’25	–	–	–	56.3
DBM-GML [40]	AAAI’25	–	–	–	57.4
CBTS [46]	TMM’25	<u>71.0</u>	58.4	<u>39.3</u>	<u>60.6</u>
GLMC+SPS	Ours	66.5	53.7	32.0	55.4
BCL+SPS	Ours	68.3	55.8	38.1	58.2
CBTS+SPS	Ours	71.2	<u>58.3</u>	39.5	60.7

2) *Detailed Analysis on ImageNet-LT*: To further dissect the source of our method’s success on this large-scale benchmark, we provide a detailed performance breakdown across Head, Medium, and Tail class subsets for ImageNet-LT in Table IV.

First, our method brings substantial and consistent improvements to tail-class performance. When combined with GLMC and BCL, SPS boosts tail accuracy by +2.2% and +1.6%, respectively. This supports our core hypothesis: by synthesizing high-quality guidance and transferring distilled semantic knowledge, we can effectively enrich the feature representations. Moreover, our framework shows favorable behavior on non-tail classes. Furthermore, SPS demonstrates its ability to push the envelope even further, achieving a new state-of-the-art tail accuracy of 39.5% when combined with CBTS, underscoring its continued efficacy in the most advanced method. While smaller datasets like CIFAR-LT exhibit slight medium-class drops due to strong feature reorganization, ImageNet-LT reveals a different trend—performance on head and medium classes remains stable or even improves. We attribute this to the same mechanism acting at different scales: on large, sparse spaces, semantic guidance serves as a global regularizer that benefits the overall structure.

D. Ablation Study

TABLE V: Ablation study of SPS components and fusion strategies on CIFAR-100-LT (IR=100). GLMC is used as the baseline. Accuracy (%) is reported for head, medium, tail, and overall classes. Fusion strategy: A = all-head fusion, R = random-head fusion, V = visual-similarity fusion, C = CLIP-guided fusion, and – = no fusion.

ProtoCL	SemLRF	Fus.	ACC	Head	Mid	Tail
×	×	–	58.20	73.90	60.40	37.90
✓	×	–	58.50	73.40	59.70	40.50
×	✓	A	58.56	73.29	59.68	40.39
×	✓	R	58.43	72.94	59.88	40.45
×	✓	V	58.35	72.17	59.68	41.29
×	✓	C	59.00	73.60	60.20	41.10
✓	✓	C	59.00	74.10	58.70	42.50

1) *Module Ablation*: To rigorously evaluate the contribution of each SPS component, we conduct an ablation study on CIFAR-100-LT (IR=100) by incrementally adding Semantic Low-Rank Fusion (SemLRF) and Prototype Contrastive Loss (ProtoCL) to the GLMC baseline, as shown in Table V. For the fusion-strategy comparison, all-head fusion uses all head classes as transfer sources, random-head fusion randomly selects five head classes to match the default neighbor number, and visual-similarity fusion selects the five visually nearest head classes according to prototype similarity. Compared with the baseline, ProtoCL alone improves tail accuracy from 37.90% to 40.50%, indicating that prototype-level contrastive regularization enhances global class separation. SemLRF with CLIP-guided fusion further improves tail accuracy to 41.10% and overall accuracy to 59.00%, confirming that semantically guided prototype synthesis effectively enriches tail-class representations.

We further compare CLIP-guided fusion with all-head, random-head, and visual-similarity fusion to examine the importance of source selection. Although all-head and random-head fusion improve tail accuracy over the baseline, their tail performance remains comparable to ProtoCL alone, suggesting that simply increasing the number of transferred head classes or randomly selecting sources does not provide sufficiently reliable knowledge for tail-class enhancement. Visual-similarity fusion achieves a higher tail accuracy of 41.29%, but it substantially reduces head accuracy from 73.90% to 72.17%, likely because prototype similarity is computed in an imbalanced visual feature space and may introduce biased local appearance correlations. In contrast, CLIP-guided fusion achieves a better balance between tail-class improvement and head-class preservation, reaching 41.10% tail accuracy and 59.00% overall accuracy with a much smaller head-class drop. When combined with ProtoCL, the full SPS further improves tail accuracy to 42.50% and head accuracy to 74.10%, demonstrating that semantic source selection, low-rank purification, and prototype-level separation are complementary.

2) *Analysis of λ_s and λ_p* : First, as illustrated in Figure 4, our method shows strong stability, with overall accuracy consistently ranging between 58.0% and 59.0% across different hyperparameter settings. This suggests robustness to hyperparameter choices without requiring fine-grained tuning. Next,

increasing the SemLRF weight (λ_s) with $\lambda_p = 2$ improves performance, peaking at 59.0% for $\lambda_s = 10$, validating the importance of semantic fusion. However, setting $\lambda_s = 20$ slightly reduces performance, suggesting that excessive emphasis on fusion may compromise classification and hinder training stability. Similarly, varying the ProtoCL weight (λ_p) with $\lambda_s = 10$ yields peak accuracy at $\lambda_p = 2$, beyond which performance declines. This implies that while ProtoCL helps enforce inter-class separation, too much regularization may disturb head-class representations. Overall, the setting $\lambda_s = 10$, $\lambda_p = 2$ strikes an optimal balance, enhancing tail-class prototypes via SemLRF while regularizing global structure through ProtoCL.

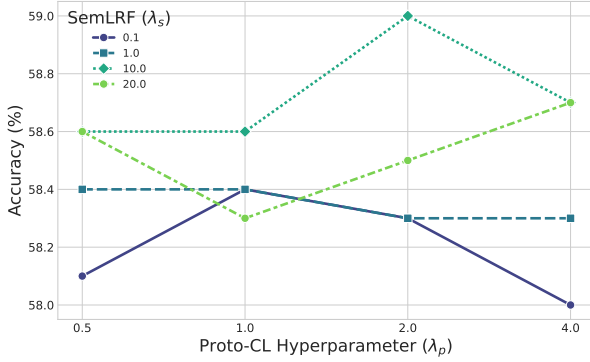


Fig. 4: Sensitivity to loss weights λ_s and λ_p .

E. Visualization of module

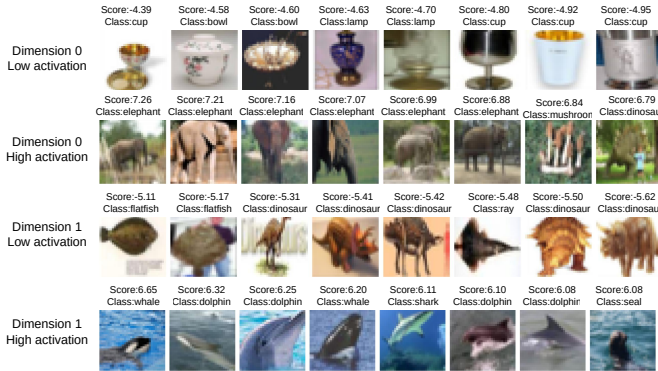


Fig. 5: Qualitative analysis of the top two principal dimensions (SVD 0 and 1) extracted from head-class prototypes on the CIFAR-100-LT dataset.

1) *Visualization Analysis of SVD Dimensions*: To qualitatively assess whether our method extracts meaningful and interpretable concepts from head-class prototypes, we analyze the principal semantic dimensions learned on the CIFAR-100-LT dataset (IR=100). First, as illustrated in Figure 5, our analysis reveals that the most significant dimension, SVD Dimension 0, effectively captures a fundamental axis distinguishing natural creatures from man-made artifacts. High-activation images primarily include large, textured animals like “elephant”, while low-activation ones correspond to smooth, small objects like “cup” and “bowl”. Next, SVD Dimension

1 further refines semantic distinctions, separating aquatic, streamlined animals from terrestrial or benthic creatures. High-activation images include marine mammals like “dolphin” and “whale”, while low-activation ones feature “dinosaur” and “flatfish”. Finally, these results demonstrate that our method not only compresses information but also extracts high-level, transferable semantic concepts independent of specific class identities.

F. Analysis of Class Separability

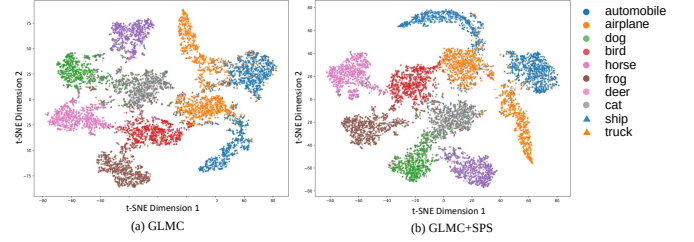


Fig. 6: T-SNE plot comparing the feature spaces learned by GLMC and GLMC+SPS on CIFAR-10-LT (IR=100). The circle represents the head-class, and the triangle represents the tail-class.

1) *Qualitative Analysis*: To validate the efficacy of our framework, we conduct a comparative t-SNE analysis of the learned feature space on CIFAR-10-LT (IR=100). First, as shown in Figure 6a, the baseline feature space is poorly organized, tail-class samples are scattered and intermixed with head classes. This reflects weak intra-class compactness and low inter-class separability—common issues in long-tailed learning. In contrast, our method (Figure 6b) yields well-clustered features and clearly separated class boundaries. These improvements stem from the synergy of SemLRF and ProtoCL: SemLRF transfers semantic knowledge to refine tail-class prototypes, while ProtoCL enhances prototype-level discrimination. Finally, this visualization qualitatively confirms that SPS not only improves classification but also leads to a more structured and semantically meaningful feature space.

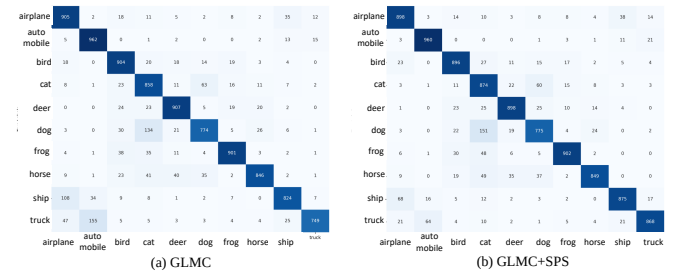


Fig. 7: Confusion matrix comparing the GLMC with GLMC+SPS on the CIFAR-10-LT (IR=100) dataset.

2) *Confusion Matrix Analysis*: To further dissect the model’s performance at a granular level, we present a comparative analysis of the confusion matrices for the baseline (GLMC) and our SPS model, evaluated on the CIFAR-10-LT (IR=100) dataset (Figure 7). First, the confusion matrix of the baseline model (Figure 7a) reveals difficulty in classifying

tail classes, with prominent off-diagonal entries indicating misclassifications. For instance, “truck” samples (a tail class) are often confused with “automobile” (a head class), reflecting a bias toward well-represented classes under uncertainty. In contrast, the SPS model (Figure 7b) shows stronger diagonal dominance, especially for tail classes, indicating improved classification. The “truck” class, for example, exhibits significantly fewer misclassifications and higher true positive rates, demonstrating enhanced feature discriminability. Crucially, these gains do not come at the cost of head-class performance. Diagonal entries for head classes such as “airplane” and “automobile” remain largely unchanged, showing that SPS improves tail-class accuracy without inducing catastrophic forgetting on head classes.

G. Intra-class Compactness Evaluation

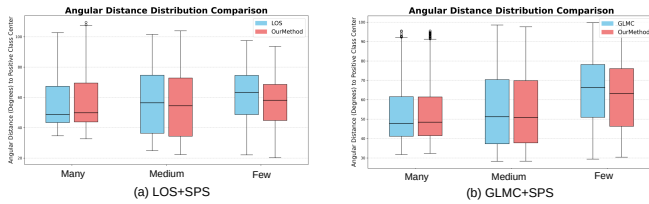


Fig. 8: Comparison of angular distance distribution of sample features to their class prototypes.

To quantitatively evaluate the impact of our method on the learned feature space, we analyze the angular distance between test features and their corresponding class prototypes on the CIFAR-100-LT (IR=100) dataset. First, as shown in Figure 8, our method notably improves tail-class feature quality. Compared to the baseline (blue), our model (red) yields lower median angular distances and tighter distributions, indicating enhanced intra-class compactness and reduced feature dispersion. This demonstrates the effectiveness of Semantic Low-Rank Fusion (SemLRF) in transferring semantically relevant knowledge to tail classes. Second, in the ‘Many’ group, a slight increase in median distance reflects a controlled regularization effect rather than degraded performance. The combination of low-rank constraints and contrastive prototype learning mitigates overfitting and encourages generalizable feature representations. Final classification results confirm that head-class accuracy remains stable, validating the balance achieved by our representational strategy.

V. CONCLUSION

We propose a Semantically-Guided Prototype Synthesis (SPS) framework to tackle the “purity–breadth dilemma” in long-tailed recognition. SPS combines Semantic Low-Rank Fusion Loss (SemLRF) and Prototype Contrastive Loss (ProtoCL) to enable effective knowledge transfer and enhance feature discrimination. Specifically, SemLRF constructs a purified visual subspace via singular value decomposition of head-class prototypes, capturing transferable knowledge and suppressing class-specific noise. Semantically matched head-tail pairs, generated using a pre-trained CLIP model, are

fused within this subspace to synthesize guidance prototypes for tail classes. ProtoCL further enhances global discrimination through contrastive learning at the prototype level. Experiments show SPS significantly boosts tail-class accuracy without harming head-class performance.

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